

IO SKY

Scheduling Agent

Functional Specification

Autonomous operational scheduling capability

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IMPLEMENTATION STANDARD

Complete, explicit, security-conscious, practical and future-ready. This specification defines launch behaviour and architecture so the Scheduling Agent can remove routine scheduling and associated administrative work while preserving human control, data protection, availability integrity, auditability and an organised customer lifecycle.

Brand operating principles: Calm · Control · Confidence · Clarity.

This document is authoritative for the Scheduling Agent and its integrations. Where it interfaces with the AI Receptionist, IVR/Telephony, AI Sales Outbound, AI Scan, Client Portal, Notification Center, permissions or platform administration, the shared platform security model remains authoritative.

1. Purpose and non-negotiable outcome

The Scheduling Agent is not a calendar widget and not a standalone sales agent. It is IO SKY's authoritative autonomous scheduling capability. It must coordinate appointments across authorised channels and agents, execute approved scheduling work, and complete the associated administration so that human users do not have to repeat routine operational steps.

CORE PRINCIPLE

A completed Scheduling Agent action must complete the associated authorised administrative workflow, not merely create a calendar event.

The intended outcome is that authorised IO SKY users can delegate scheduling work, review outcomes and focus their time on revenue generation, customer conversations, approvals and high-value decisions.

2. Scope and launch appointment types

- Discovery Call - launch-ready.
- AI Scan Review Session - launch-ready for Operations, Cyber and Elite, with product-specific rules where required.
- Custom / Manual Appointment - launch-ready when initiated by an authorised IO SKY user.
- Project Meeting, Technical Session and Support Session - architecture-ready and configurable without rebuilding the scheduling core.

A Technical Session or Support Session means the Scheduling Agent schedules the appropriate human meeting. It does not become a technical expert or support engineer.

3. Shared-agent architecture

All authorised channels and agents must use one authoritative Scheduling Engine. No channel may implement its own conflicting booking logic.

Initiator / channel	Scheduling role
AI Receptionist	Uses Scheduling Agent actions during an inbound voice conversation.
IVR fallback	Can route a caller who wants an appointment to the Scheduling Agent when the AI Receptionist is unavailable and the scheduling voice capability remains available.
Discovery Page	Provides an AI-guided scheduling journey using the same engine; not an independent Calendly-style slot picker.
Client Portal	Uses the same engine for authorised client appointment actions.
Super Admin	Can create a manual Scheduling Task and delegate the contact and booking workflow.
AI Sales Outbound	Launch integration: can invoke the same scheduling capability or create an authorised Scheduling Task from an outbound prospect interaction.
Future authorised agents	May invoke the same controlled scheduling action layer without duplicating booking logic.

4. Authoritative booking engine and zero double-booking

ABSOLUTE REQUIREMENT

A confirmed single-capacity host/resource and time combination may never be double-booked - including simultaneous calls, different channels, retries, webhook duplication, calendar sync delay, Super Admin activity

or race conditions.

- Availability must be revalidated immediately before the consequential booking action.
- Temporary holds may be used where required; a hold is never presented as a confirmed booking and expires automatically.
- Booking must be atomic. If two sessions compete for the same final slot, only one can succeed; the other receives fresh real availability.
- Rescheduling must be atomic: the old appointment remains intact until the new slot has been secured successfully.
- Server/database enforcement is required; frontend-only conflict checks are insufficient.

5. Availability model

Effective availability is calculated from appointment-type rules, working hours, host availability, external calendar free/busy state, existing IO SKY bookings, preparation time, buffers, manual blocks, exceptions, holidays/time off and required-resource availability.

- One central Scheduling Control Center / master scheduling view must show operational availability and appointments.
- External calendar details are not required for ordinary availability; free/busy and time boundaries are the default minimum.
- A caller is never told why a time is unavailable.
- Super Admin can create manual unavailable blocks without providing a reason.
- Multiple calendars per host may block availability.
- Recurring availability and one-off exceptions are supported.
- External calendar busy periods count as occupied.

6. Discovery Call launch configuration

Launch baseline: Europe/Amsterdam; phone call by default; IO SKY calls the submitted number. Discovery appointment configuration remains administratively editable.

- Meeting duration: 30 minutes.
- Protected preparation time: default 30 minutes.
- Post-meeting operational buffer: configurable 0-15 minutes; default 15 minutes for recovery/overrun, not for manual CRM administration.
- Booking horizon: default maximum 30 days ahead.
- Minimum booking lead time: default 2 hours.
- Late-cancellation threshold: 24 hours.
- Same-day booking is allowed when lead time, preparation and availability rules are satisfied.

Preparation Time, Meeting Duration and Post-Meeting Buffer are separate concepts and separate configuration values. The capacity engine must calculate actual available capacity; it must not assume a fixed number of calls per day.

7. AI Scan Review Sessions

AI Scan Review Sessions are a launch appointment type. Operations, Cyber and Elite may use product-specific duration, preparation, availability, eligibility, horizon and lead-time rules. Scheduling does not replace the internal human review/approval/publication lifecycle.

- The Scheduling Agent may schedule a customer-facing review only when the relevant Scan entitlement/workflow allows it or an authorised IO SKY user makes it available.
- The Scheduling Agent may not give away a paid or non-included review session.

- The assigned reviewer receives only authorised Scan context required to prepare for the meeting.
- Scheduling itself does not analyse the full confidential Scan dossier when that is unnecessary to perform the scheduling task.

8. Manual Scheduling Task - launch requirement

CRITICAL LAUNCH WORKFLOW

Super Admin must be able to instruct the Scheduling Agent to contact a person, explain why IO SKY is contacting them, find a suitable time and complete the appointment workflow without requiring the human user to perform each operational step.

This applies both to a person already known in IO SKY's CRM/lifecycle and to a newly encountered prospect, including a prospect first spoken to during an outbound call.

Input	Requirement
Person	Select an existing Person/Lead/Organisation or create a new person record.
Contact data	At minimum the authorised phone number and/or email; organisation where known.
Purpose	Why the Scheduling Agent is contacting the person.
Appointment objective/type	For example Discovery Call, AI Scan Review or authorised custom appointment.
Context	Relevant prior conversation/context so the agent can explain the reason naturally and accurately.
Contact timing	Start now, a specific date/time, or an authorised contact window.
Constraints	Desired period, exclusions, host requirements or other scheduling constraints.
Required outcome	What the agent must return with: ideally a confirmed appointment, or a clear documented outcome/follow-up state.

Natural-language instructions must be supported. Example: an IO SKY user can state that a prospect was just spoken to, wants a Discovery Call but does not yet know a suitable date, agreed to be contacted on Friday, and the Scheduling Agent should call on Friday, explain the purpose, find a suitable time and complete the booking.

The Scheduling Task itself must be a durable, searchable administrative record linked to the correct person/organisation before contact begins.

9. Scheduling Task lifecycle and outbound follow-up

Minimum lifecycle:

Draft -> Ready/Scheduled -> Contacting -> Reached -> Scheduling -> Booked

Alternative outcomes include No Answer, Busy, Follow-up Required, Failed and Cancelled.

- No Answer is never a dead-end status. The system records what happened and what authorised next action was taken.
- Voice, email and SMS are launch-supported scheduling channels.
- The contact strategy must be configurable and bounded; no endless autonomous calling or messaging loops.
- A prospect may request 'call me tomorrow'; the Scheduling Agent may schedule that follow-up within the same authorised task without new manual approval.
- If the prospect asks to continue by email or SMS, the agent may switch to an authorised, already provided/confirmed channel and continue the scheduling journey.
- Every attempt, channel, outcome and next step is administratively recorded.

Example operational state: Attempt = No Answer; Channel = Voice; Follow-up = Email sent; Next attempt = scheduled according to policy; or Follow-up Required = human attention.

10. Scheduling Agent is not AI Sales Outbound

A manually initiated Scheduling Task is not permission for unrestricted prospecting. The Scheduling Agent contacts a person only from an authorised scheduling task/workflow with a legitimate scheduling purpose. AI Sales Outbound is a separate launch capability, but when it needs to schedule or delegate scheduling it must use this same Scheduling Engine.

11. Natural-language scheduling intelligence

- Understand requests such as 'next Tuesday afternoon', 'preferably after three', 'as early as possible', 'not Wednesday', or 'toward the end of next week'.
- Translate natural language into scheduling constraints, then query real authoritative availability.
- Ask one targeted clarification when ambiguity materially affects the result.
- Normally present 2-3 relevant real options at a time, especially in voice interactions.
- If no suitable slot exists, intelligently search adjacent permitted periods rather than immediately declaring no availability.
- Never invent or promise a time without a live authoritative availability check.

The agent remains a scheduling specialist. It does not independently become a Sales Advisor, AI Scan consultant, Support Engineer or Technical Advisor.

12. Professional response outside scheduling expertise

If a person asks a substantive question while scheduling, the agent may answer only reliable, approved information within its governed knowledge. It must not improvise specialist advice. Where specialist input is required, it must explain this professionally, arrange the appropriate handoff/follow-up where authorised, and then continue the scheduling objective where sensible.

Situation	Required behaviour
Simple approved appointment/company question	Answer concisely from governed knowledge and continue scheduling.
Technical or specialist question beyond authority	Acknowledge the question; do not guess; arrange an appropriate expert/handoff/follow-up and continue scheduling where possible.
AI Scan substantive interpretation	Do not perform Scan consultancy; route to the authorised Scan/expert workflow.
Privacy/security concern	Minimise data, avoid legal/security conclusions, route appropriately.
Commercial promise/discount/custom pricing	Do not invent or commit beyond approved rules.

13. Data collection and reuse

Discovery Call baseline data: full name, work email, phone, company where applicable, relevant business context/reason and appointment data. Each appointment type may define its own required/optional fields.

- Do not ask again for reliable information already collected in the same authorised journey unless necessary.
- Known context improves convenience but is not identity verification.
- Caller ID may help locate a possible record but is not verification.
- Do not collect unnecessary sensitive data such as date of birth merely to secure ordinary booking changes.

14. Account creation boundary

PUBLIC REGISTRATION RULE

A Discovery Call, Scheduling Task, contact form, lead or booking must never automatically create platform credentials.

- The only public self-service account creation path is after a backend-verified successful AI Scan payment/entitlement, before the customer continues with the AI Scan journey.
- Otherwise account access exists only through an authorised IO SKY invitation.
- This invitation principle applies to Client, Developer, Technical Operator and Admin accounts.
- Super Admin must never be creatable through public self-registration.

15. Booking confirmation flow

1. The agent identifies the intended appointment type and constraints.
2. The authoritative engine retrieves live availability.
3. The person selects an option.
4. The agent summarises the appointment type, date/time and relevant method/timezone and asks for confirmation.
5. The backend atomically revalidates and commits the booking.
6. Only after backend success may the agent say the appointment is booked.
7. The system creates the unique booking reference and all downstream administrative events.

NO FALSE SUCCESS

The AI, frontend or integration may never infer a successful booking from intent or an intermediate state. Only authoritative backend confirmation permits a success statement.

16. Booking reference and global searchability

Every confirmed appointment receives a unique, non-easily-predictable, internally searchable booking reference. The reference helps locate a booking but is not, by itself, sufficient authorisation for sensitive actions.

The platform must provide permission-aware global search across relevant operational identifiers, including booking reference, name, organisation, email, phone and relevant platform identifiers.

Search filters must work individually and in combinations. Examples include Appointment Type + Status + Host + Date Range; Organisation + AI Scan; Source = AI Sales Outbound + Follow-up Required.

OPERATIONAL ORGANISATION PRINCIPLE

One person/organisation, one coherent operational lifecycle - searchable, linked, auditable and permission-aware. Every automated workflow must leave IO SKY's administration more complete and organised, never more fragmented.

17. 360-degree operational lifecycle

Where authorised, a user who finds a person or organisation should be able to navigate the coherent relevant lifecycle: prospect/lead, contact interactions, calls and structured summaries, Scheduling Tasks, bookings, Discovery Calls, AI Scan records, client relationship, projects, relevant documents, communications, actions and current follow-up state.

- Agents must not create disconnected shadow records when an existing authorised entity should be linked.
- Potential duplicate matches must not be blindly merged.

- Searchability never expands authorisation.
- Technical retries, webhooks and provider errors belong in operational/audit logs, not as noisy customer-lifecycle events.

18. Timezones and date handling

- Canonical launch timezone: Europe/Amsterdam.
- International prospects may be shown times in their chosen/reliably known local timezone while backend data remains timezone-safe.
- When timezone is ambiguous, explicitly confirm it rather than guessing.
- Use timezone-aware date/time handling and real DST rules; never hardcode UTC+1/UTC+2 offsets.

19. Rescheduling

Rescheduling requires sufficient verification and explicit confirmation. The old appointment must remain valid until the new slot is successfully committed.

- Validate and secure the new slot first.
- Perform the move atomically.
- Release the old slot only after success.
- If the new booking fails, explicitly tell the person that the existing appointment remains in the system at its original date/time.
- Offer a safe next step: search another slot, retry, use the authorised portal route where applicable, or arrange human follow-up.
- Send an updated confirmation after success and update calendar, lifecycle, notifications and audit consistently.

20. Telephone verification for sensitive booking changes

8. Locate the booking using booking reference or sufficient lookup data.
9. Offer a one-time verification code to an already linked/registered email address or phone number. If both are available, let the person choose email or SMS.
10. Do not unnecessarily read the full destination aloud; use masking where appropriate.
11. Use a short numeric OTP, default six digits, single-use, short-lived, rate-limited and with limited attempts.
12. During a voice call, prefer DTMF entry where supported so the code need not be spoken and unnecessarily captured in audio/transcription.
13. Only the server can set verification success.
14. Only after successful verification may the consequential reschedule/cancellation action proceed.

A booking reference, name, email address, caller ID or other lookup knowledge is not sufficient verification by itself.

21. Portal verification

For an authenticated portal user, a valid authenticated session plus backend-confirmed ownership/authorisation and appropriate security state may authorise ordinary booking actions. Step-up authentication may be required when risk or security policy demands it. The portal must not require redundant OTP for every normal action when existing authentication is sufficient.

22. Cancellation and late cancellation

- Cancellation requires sufficient verification and explicit final confirmation.
- Discovery Call late-cancellation threshold: 24 hours before start.
- Cancellation remains technically possible inside the threshold, but is recorded as Late Cancellation.

- No penalty, fee or other commercial sanction is implied unless separately approved and legally/commercially implemented.
- A released slot is recalculated against current lead-time, preparation, host and availability rules and may become available again when valid.
- Cancelled, Late Cancellation and No-show are distinct lifecycle/analytics states.

23. Appointment lifecycle

Minimum business states: Pending/Held -> Confirmed -> Rescheduled and/or Cancelled/Late Cancellation -> Completed or No-show. Technical delivery/sync states must remain separate from business appointment status.

No-show follow-up may be configured, but the system must not silently rebook a person without confirmation.

24. Confirmation and reminder communications

- Automatic confirmation immediately after a successful booking.
- Discovery default reminders: 24 hours before and 1 hour before; configurable per appointment type.
- Launch channels for scheduling workflows: email, SMS and voice where appropriate and authorised.
- Confirmation/reminder content includes appointment type, date/time, timezone where relevant, duration, method/location, booking reference where appropriate, necessary instructions and a safe modification/cancellation route.
- Delivery must be idempotent; retries must not create duplicate messages.

25. Professional multilingual behaviour

LANGUAGE QUALITY REQUIREMENT

Launch scheduling interactions must support professional Dutch and English. Approved meaning, business rules, certainty, conditions and context must remain equivalent across languages. Machine translation must not be used as a substitute for professional localisation.

Where wording is mandatory/canonical - including verification, confirmation, privacy/security notices, cancellation/reschedule language and other fixed communications - developers must implement approved Dutch and English formulations rather than loosely interpreting one language from the other.

Natural conversation does not require awkward word-for-word translation, but it must communicate the same approved meaning and must never add, omit or alter business rules.

The same principle applies platform-wide to website and platform copy: every supported language must professionally convey the exact same approved content and context as the English source. Do not use raw machine translation that changes nuance, scope, commercial meaning, legal meaning or context. Additional languages require professional-quality localisation and validation before publication.

26. Preparation and briefing

- Preparation Time is a real protected scheduling block and prevents overlapping appointments for the required host.
- The relevant briefing must be ready before the preparation period begins.
- Discovery briefing should include, where authorised: person, company, contact details, reason, relevant business context, known needs/pain points, relevant prior communication, booking source, desired outcome and key unknowns.
- AI Scan Review briefing is Scan-specific and must expose only authorised information.
- The Scheduling Agent links the correct records/context; a dedicated Intelligence/Briefing capability creates the substantive preparation brief.

27. Automated post-call administration

IO SKY must not require a human to spend routine time re-entering information after each meeting. Where the relevant data and permissions exist, completion of a call/meeting should trigger authorised automated administration.

Typical flow: Call/meeting completed -> structured summary -> lifecycle/timeline update -> agreed actions extracted -> follow-up/task state -> prospect/client record update -> analytics -> next authorised workflow/notification.

If transcription/summarisation is enabled and legally/configurationaly permitted, it should support this automation. The durable business artefact should be a factual structured summary rather than unnecessary permanent raw transcript storage, consistent with the telephony/privacy specifications.

28. Notifications

- New relevant appointment -> immediate Notification Center event for the authorised assignee.
- Reschedule -> one clear lifecycle change showing old time -> new time, not two apparently active appointments.
- Cancellation, Late Cancellation and No-show -> correct actionable state/event where relevant.
- Preparation -> a concise 'briefing ready' notification aligned with the preparation period; avoid notification spam.
- Manual Scheduling Task outcomes and failures -> show attempt, channel, outcome, next step and human attention requirement where applicable.

29. Calendar source of truth and sync

- IO SKY Booking Engine/database is authoritative for IO SKY appointments.
- External calendars contribute free/busy availability and receive synchronisation.
- Deleting an external calendar event does not automatically cancel the IO SKY booking.
- Calendar sync state must be visible, at minimum Pending, Synced and Failed/Retry Required.
- A calendar provider outage does not delete or invalidate the authoritative IO SKY booking.
- Persistent sync discrepancies/failures must retry safely and surface operationally.

30. Multi-host architecture - launch foundation

NOT A FUTURE RETROFIT

The scheduling architecture must be multi-host from the initial implementation. IO SKY expects multiple employees in the near term; the system must not be hardcoded around one founder, one calendar or one host.

- Each host has own availability, calendars, manual blocks, exceptions/time off and eligible appointment types.
- Appointment types define eligible hosts.
- Host strategies must support or be architecturally ready for specific host, first available, round-robin, priority and role/skill eligibility.
- Super Admin can reassign an appointment only to an eligible and actually available host, with correct calendar, ownership, briefing/access and notification updates.
- Appointments may require multiple IO SKY participants from the beginning; the availability engine finds a time when all mandatory resources are available.
- Appointment types may support multiple external attendees.

31. Capacity and group semantics

Standard Discovery is one prospect/company appointment against the required host capacity. The architecture may support future group/multi-capacity appointment types, but single-capacity resources may not be double-booked.

32. Booking policy and waitlist

- Booking horizon and minimum lead time are configurable per appointment type.
- Discovery defaults: 30-day horizon and 2-hour minimum lead time.
- AI Scan Review uses its own configurable rules.
- Waitlist capability may be architecture-ready but launch-disabled.
- A waitlist must never silently move or create a booking: available slot -> offer -> person accepts -> authoritative booking.

33. Duplicate and abuse protection

- Before creating a new future appointment, detect whether the same person/contact already has a relevant future appointment and clarify whether the intent is to reschedule or create an additional legitimate appointment.
- Do not impose a simplistic one-person/one-appointment rule; multiple appointments can be legitimate.
- Public scheduling actions require server-side input validation, rate limiting, duplicate protection, monitoring and proportionate bot/abuse controls.

34. Super Admin Scheduling Control Center

Super Admin must be able to operate the Scheduling system without routine developer intervention.

- View day/week/month scheduling, bookings, preparation, buffers, manual blocks, external busy state, hosts, statuses and sync/failure state.
- Create and manage appointment types, duration, preparation, buffers, availability, booking horizon, lead time, hosts, eligibility, meeting method/location, reminders, fields, cancellation/reschedule policies, calendars, exceptions/holidays and Scheduling Task contact rules.
- Create Manual Scheduling Tasks for existing or new prospects and provide normal-language context/instructions.
- Override ordinary availability with explicit warning, confirmation and audit.
- Do not allow accidental double-booking of a single-capacity host merely because the user is Super Admin.

35. Configuration lifecycle

Material structural configuration should support Draft -> Test/Preview -> Authorised Publish -> Active -> Version History/Rollback. A simple immediate availability block such as 'unavailable tomorrow afternoon' does not require this heavy publication flow.

36. Canonical scheduling permissions

The implementation must support granular permissions at least equivalent to the following canonical capability set:

Permission	Purpose
booking.view	View authorised bookings.
booking.create	Create bookings.
booking.reschedule	Reschedule authorised bookings.

booking.cancel	Cancel authorised bookings.
booking.override_availability	Perform an explicitly authorised availability override.
availability.view	View authorised availability.
availability.manage	Manage authorised availability/blocks.
appointment_type.view	View appointment type configuration.
appointment_type.manage	Manage appointment type configuration.
scheduling_task.view	View authorised Scheduling Tasks.
scheduling_task.create	Create a Scheduling Task.
scheduling_task.manage	Manage/retry/cancel authorised Scheduling Tasks.
scheduling_host.view	View authorised host configuration.
scheduling_host.manage	Manage host eligibility/configuration.
scheduling_integration.manage	Manage authorised calendar/communication integrations.
scheduling_communication.manage	Manage authorised scheduling communication/channel rules.
scheduling_audit.view	View scheduling audit records.
scheduling_analytics.view	View scheduling analytics.
scheduling_configuration.manage	Manage broader Scheduling configuration.

Exact internal identifiers may be mapped technically, but the capabilities must remain separately delegable and must not be collapsed into a single broad 'Scheduling Admin' permission.

37. Role, scope and Technical Operator isolation

UNIVERSAL AUTHORISATION RULE

Role + Permission + Scope + Assignment/Ownership + Environment + Resource determine access. Search, UI visibility, AI context or notification never grants access.

- Super Admin: full authorised Scheduling Control Center.
- Admin: only explicitly delegated permissions and scope; cannot self-expand access.
- Client: only own authorised appointments and permitted actions.
- Developer: no prospect/client scheduling data by default; only assigned development access.
- Technical Operator: only the exact technical scopes/resources explicitly granted.

Technical Operator scopes must be easy for Super Admin to select individually or in combination, including Frontend, Backend, Full Stack, UI/UX, Infrastructure/DevOps, Security, Database/Data, specific portal(s) and specific AI capability/agent. Granting access to one AI or one portal must not grant visibility across the whole platform.

Customer and IO SKY data must remain protected at all times. Sensitive production data access must be necessary, explicit, minimal and auditable.

38. Security boundaries

- Booking/session state is server-side authoritative.
- Booking references must not be predictable sequential identifiers when exposed externally.
- OTP is single-use, expires, has limited attempts and rate limits.
- The AI/frontend cannot fabricate verification state.
- Caller ID, name, email or booking reference may locate a record but do not independently verify identity.
- Consequential actions require confirmation immediately before execution.
- The agent cannot change business-hour, availability, permission or commercial policy rules through conversation.

39. Failure handling

Failure	Required behaviour
Availability source unavailable	Do not say 'nothing is available'. State that availability cannot

	currently be checked reliably and offer an appropriate fallback.
Booking commit fails	Never claim success. Retry safely where appropriate or offer another slot/human follow-up.
Reschedule fails	Explicitly state that the original appointment remains intact.
Confirmation email/SMS fails	Booking remains valid; delivery state retries and becomes operationally visible.
Calendar sync fails	Authoritative booking remains valid; retry/alert.
Outbound voice no answer/busy	Record distinct outcome and execute the bounded authorised follow-up policy.
Provider/technical failure	Preserve state, log the exact failed step, avoid false success and use safe fallback.

Partial failures must be distinguishable by step; a generic Error state is insufficient for operational support.

40. Idempotency and retry safety

Retries, webhooks, provider redelivery and user retries must not create duplicate bookings, duplicate reminders, duplicate Scheduling Tasks or duplicate lifecycle events. Consequential operations require idempotency protection and deterministic reconciliation.

41. Concurrency and session isolation

10 IS A TEST CASE, NOT A LIMIT

The system must pass at least ten genuinely simultaneous differentiated scheduling journeys, but ten must never be implemented or interpreted as maximum capacity.

- Each concurrent session is fully isolated: identity, OTP, selected slot, context, booking state, tool results and communications cannot leak between sessions.
- Architecture must scale materially beyond ten without redesigning the core session/booking model.
- Load tests must include competing requests for the same slot and multiple different appointment journeys.
- Capacity limits of providers/accounts must be documented, monitored and provisioned appropriately.

42. AI Receptionist and IVR integration

- Normal inbound path: caller -> AI Receptionist -> Scheduling Agent actions when scheduling is needed.
- If AI Receptionist is unavailable but telephony/IVR and Scheduling Agent voice capability remain available, IVR must provide an appropriate route for callers who want to schedule.
- The Scheduling Agent then gathers the necessary scheduling context and completes the authorised scheduling journey.
- If the Scheduling Agent is also unavailable, IVR must not offer a false working scheduling path; use controlled callback/contact fallback.
- The dependency/failure matrix must make clear which telephony/AI components can fail independently.

43. Discovery Page behaviour

The Discovery Page may provide a guided scheduling journey, but it must use the same authoritative Scheduling Engine and business rules. It must not become a separate self-service calendar implementation with divergent availability or conflict logic.

44. AI Sales Outbound integration - launch

AI Sales Outbound is a launch source/integration, not a future placeholder. During an authorised outbound prospect interaction, it may invoke the same Scheduling Engine or create an authorised Scheduling Task. If a prospect says they cannot choose a date now and asks IO SKY to contact them later, the agreed follow-up can become a Scheduling Task with the relevant context, contact time/window and objective.

AI Sales Outbound must not implement its own duplicate booking, availability or confirmation logic.

45. Communication timeline and source attribution

- Booking and Scheduling Task events link to the unified Person/Lead/Organisation/Client lifecycle where safely matched.
- Source/channel attribution includes AI Receptionist, Scheduling Agent, Discovery Page, Client Portal, Super Admin and AI Sales Outbound.
- Relevant lifecycle events include booking created, rescheduled, cancelled/late-cancelled, completed and no-show.
- Technical retry noise belongs in operational/audit logs, not customer-facing lifecycle history.

46. Analytics and operational visibility

- Booking requests and successful/failed bookings.
- Appointment type and source/channel.
- Reschedules, normal cancellations, late cancellations, no-shows and completed appointments.
- Booking lead time and availability/utilisation.
- Reminder/confirmation delivery state.
- Calendar sync failures.
- Prevented booking conflicts.
- Scheduling Agent contact attempts, reached/no-answer/busy/failure outcomes and booking conversion.
- Host utilisation where appropriate.
- Current/peak concurrency and capacity/saturation events where available.

Do not invent a fake AI performance score. Analytics should be factual and operational.

47. Privacy and data minimisation

- Use only the context necessary for the scheduling purpose and authorised downstream administration.
- Do not expose external calendar event titles/details to callers when availability is sufficient.
- Do not place unnecessary sensitive content in email/SMS notifications.
- Do not retrieve an entire customer history merely because it exists; retrieve relevant authorised context.
- Persistent transcripts are not the default scheduling memory; structured factual summaries are preferred where conversation processing is enabled.

48. Professional caller experience

The caller should experience an organised, natural IO SKY process, not internal system mechanics. The Scheduling Agent should say what is useful to the person, not narrate database queries, CRM classifications, agent routing, confidence scores or internal state.

Examples must be treated as behavioural guidance unless explicitly marked canonical/mandatory. The agent must not rigidly recite scripts when natural contextual language is more appropriate and equally compliant.

49. Required developer acceptance scenarios

15. AI Receptionist books a Discovery Call end-to-end.
16. IVR fallback routes to Scheduling Agent when Receptionist is unavailable but scheduling voice is available.
17. Super Admin creates a Manual Scheduling Task for an existing CRM prospect.
18. Super Admin creates a Manual Scheduling Task for a newly encountered outbound prospect.
19. Scheduling Task contacts at a specified time/window and explains the correct purpose/context.
20. Voice no-answer outcome triggers the configured email/SMS follow-up and records the next step.
21. Prospect asks to be called tomorrow; agent schedules the follow-up within the authorised task.
22. Prospect switches from voice to email/SMS scheduling.
23. Discovery Call booking completes with unique reference, lifecycle link, calendar state, confirmation and reminders.
24. AI Scan Review Session booking respects Scan-specific rules and eligible reviewer.
25. Two simultaneous sessions compete for the same slot; only one succeeds.
26. At least ten genuinely simultaneous differentiated scheduling journeys remain isolated.
27. External calendar busy event blocks a conflicting slot.
28. Manual unavailable block prevents booking without exposing the reason.
29. Preparation time blocks conflicting capacity.
30. Atomic reschedule succeeds and releases the old slot only after new success.
31. Atomic reschedule fails and the original appointment remains intact; caller is explicitly told so.
32. Telephone OTP verification by registered email and SMS; DTMF entry where supported.
33. Invalid/expired/reused OTP is rejected with safe fallback.
34. Authenticated portal owner performs an authorised normal booking action.
35. Normal cancellation and Late Cancellation inside the 24-hour Discovery threshold are correctly distinguished.
36. Released late-cancel slot is recalculated against current lead/preparation rules.
37. Confirmation/reminder delivery failure does not invalidate the booking.
38. External calendar sync failure does not invalidate the booking.
39. Timezone ambiguity is clarified; DST boundary is handled correctly.
40. Super Admin availability override requires warning, confirmation and audit.
41. Single-capacity host cannot be accidentally double-booked by Super Admin.
42. Multi-host eligible-host selection works.
43. Appointment requiring multiple IO SKY participants checks joint availability.
44. Technical Operator with only one selected scope cannot access unrelated scheduling/customer/platform data.
45. Global search finds the correct authorised lifecycle by booking reference, name, organisation, email and phone.
46. Combined search filters return coherent permission-aware results.
47. AI Sales Outbound invokes the same Scheduling Engine and does not duplicate booking logic.
48. Idempotent retry/webhook processing creates no duplicate booking, reminder or lifecycle record.
49. Availability provider outage is not misrepresented as 'no availability'.
50. Scheduling analytics and source attribution reflect the actual workflow.
51. Professional Dutch and English flows preserve the same approved meaning and business rules.

50. Developer implementation evidence required

- Architecture diagram showing the authoritative Scheduling Engine and integrations with AI Receptionist, IVR, Discovery Page, Client Portal, Super Admin, AI Sales Outbound, calendar providers, communications, CRM/lifecycle, notifications and briefing trigger.
- Data/state diagram for Booking, Scheduling Task, Person/Organisation linkage, holds, verification, sync state and lifecycle.
- Evidence of atomic conflict prevention and reschedule behaviour.
- Evidence of at least ten simultaneous differentiated sessions plus capability to load-test above ten.
- Provider/account concurrency limits and the scaling/provisioning method.

- Demonstration of Manual Scheduling Task for both existing and newly created prospects.
- Demonstration of voice/email/SMS follow-up paths and bounded retry policy.
- Failure-path demonstrations: availability, booking, email/SMS, calendar sync, voice/provider and OTP.
- Permissions demonstration including granular Technical Operator scopes and specific AI/portal resource access.
- Global search/filter demonstration including combined filters and permission enforcement.
- Audit and analytics demonstration.
- Multilingual implementation evidence showing approved Dutch and English meaning parity; website/platform localisation process must also comply with the professional translation requirement.

A single happy-path demo or statement that 'tests passed' is insufficient.

51. Definition of Done

The Scheduling Agent is not complete merely because a calendar event can be created. It is complete only when the authoritative scheduling engine, launch appointment types, manual scheduling delegation, AI Receptionist/IVR/Discovery/Portal/AI Sales Outbound integrations, multi-host architecture, availability integrity, atomic booking/rescheduling, verification, confirmations/reminders, voice/email/SMS follow-up, administration/lifecycle, search/filtering, preparation/briefing trigger, post-call automation, permissions, audit, analytics, failure recovery, concurrency and acceptance evidence are implemented and passed in the intended environment.

Any missing production provider account, credential, capacity, calendar integration, communication provider, DNS/number dependency or other external prerequisite must be explicitly identified as a production prerequisite; it must not be silently omitted from the implementation.

52. Non-negotiable implementation principles

52. One authoritative Scheduling Engine for every authorised channel and agent.
53. No double-booking of single-capacity required resources.
54. No false booking, reschedule, cancellation, message-delivery or sync success.
55. Manual Scheduling Tasks are a launch capability for existing and newly encountered prospects.
56. Voice, email and SMS scheduling workflows are supported from launch.
57. AI Sales Outbound integrates with the same Scheduling Engine from launch.
58. Multi-host is part of the initial architecture, not a later retrofit.
59. A booking action completes its authorised downstream administration.
60. Global search and combined filters support an organised, coherent lifecycle while never bypassing authorisation.
61. Account creation follows the strict AI Scan payment/invitation rules.
62. Sensitive booking changes use real server-side verification.
63. Technical access is granular and never implies platform-wide/customer-data visibility.
64. Dutch, English and every future supported language preserve approved content and context through professional localisation; raw machine translation is not an acceptable publication method.
65. Every automated workflow must leave IO SKY's administration more complete and organised, never more fragmented.

53. Final architectural statement

SCHEDULING AGENT

The Scheduling Agent is an autonomous operational scheduling capability. It must remove routine scheduling and associated administrative work from authorised IO SKY users while preserving human control, security, availability integrity, contextual continuity, professional communication, auditability and a professionally organised customer lifecycle.

54. Document control

Field	Value
Document	IO SKY Scheduling Agent Functional Specification
Version	1.0
Status	FINAL FOR IMPLEMENTATION
Date	17 September 2026
Primary launch languages	Dutch and English
Canonical timezone	Europe/Amsterdam
Authoritative principle	One Scheduling Engine; no divergent booking logic.

Changes to requirements marked non-negotiable, security-sensitive, account-registration-related, verification-related, availability-integrity-related or language-equivalence-related require explicit authorised IO SKY approval before implementation changes are accepted.